

Normed Spaces and L^p Spaces

Anqiao Ouyang

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Introduction

Article I constructed measure spaces and Lebesgue measure on $\mathbb{R}^n$; Article II defined measurable functions and the Lebesgue integral on a general measure space and proved the monotone convergence theorem, Fatou's lemma and the dominated convergence theorem. The present article changes the point of view: instead of studying functions one at a time, we regard a whole class of functions as the points of a vector space and use the integral to measure the “size” of these points and the “distance” between them. The basic objects of this point of view are the L^p spaces.

L^p spaces combine three kinds of structure: a linear structure (functions can be added and multiplied by scalars), a metric structure (the norm induces a distance, so that convergence, Cauchy sequences and completeness make sense), and a measure-theoretic structure (the norm is defined by an integral). The plan is as follows. Section 1 introduces the language of normed and Banach spaces. Section 2 defines the L^p spaces and explains why functions that agree almost everywhere must be identified. Section 3 proves the inequalities of Young, Hölder and Minkowski, together with their cases of equality. Section 4 proves the completeness of L^p for $1 \leq p < \infty$ (the Riesz–Fischer theorem). Section 5 treats L^∞ separately. Section 6 compares L^p spaces with different exponents. Section 7 discusses dense subspaces and separability. Section 8 gives a brief introduction to the dual of L^p .

Conventions

- $\mathbb{K}$ denotes $\mathbb{R}$ or $\mathbb{C}$. Unless stated otherwise, $(X, \mathcal{M}, \mu)$ is a measure space, and “a.e.” (almost everywhere) refers to μ . m denotes Lebesgue measure on $\mathbb{R}^n$.
- Measurable functions, the integral of nonnegative measurable functions, and the integral of integrable functions are as in Article II. A complex-valued function $f = u + iv$ is measurable if u and v are; if $|f|$ is integrable we set $\int f \, d\mu = \int u \, d\mu + i \int v \, d\mu$. Then $|\int f \, d\mu| \leq \int |f| \, d\mu$: choose $|\alpha| = 1$ with $\alpha \int f \, d\mu = |\int f \, d\mu|$; then

$$\left| \int f \, d\mu \right| = \int \operatorname{Re}(\alpha f) \, d\mu \leq \int |f| \, d\mu.$$

- In $[0, \infty]$ we use the conventions $0 \cdot \infty = 0$, $1/\infty = 0$, $\infty^{1/p} = \infty$.
- For $1 \leq p \leq \infty$, the number $q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$ is called the **conjugate exponent** of p ; $q = \infty$ if $p = 1$ and $q = 1$ if $p = \infty$. For $1 < p < \infty$ we have $q = \frac{p}{p-1}$ and $(p-1)q = p$.

1 Normed Spaces and Banach Spaces

1.1 Norms

Article I introduced metrics on arbitrary sets. When the set is a vector space it is more natural to measure first the “length” of each vector and then to obtain distances from lengths.

DEFINITION 1.1 Norm

Let V be a vector space over $\mathbb{K}$. A map $\|\cdot\| : V \rightarrow [0, \infty)$ is a **norm** on V if for all $x, y \in V$ and $\alpha \in \mathbb{K}$:

- (i) (definiteness) $\|x\| = 0$ implies $x = 0$;
- (ii) (homogeneity) $\|\alpha x\| = |\alpha| \|x\|$;
- (iii) (triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$.

A map satisfying only (ii) and (iii) is a **seminorm**. A vector space equipped with a norm, $(V, \|\cdot\|)$, is a **normed space**.

REMARK 1.2

- (i) Homogeneity may be weakened to $\|\alpha x\| \leq |\alpha| \|x\|$ for all α : for $\alpha \neq 0$, $\|x\| = \|\alpha^{-1}(\alpha x)\| \leq |\alpha|^{-1} \|\alpha x\|$ gives the reverse inequality, and for $\alpha = 0$, $\|0\| = \|0 \cdot 0\| \leq 0$.
- (ii) The triangle inequality says that the norm of a sum does not exceed the sum of the norms, not the other way round.
- (iii) A norm is a function of a single vector; the formula $d(x, y) = \|x - y\|$ defines a metric on V : definiteness gives $d(x, y) = 0 \Leftrightarrow x = y$, homogeneity with $\alpha = -1$ gives symmetry, and the triangle inequality for the norm gives the triangle inequality for d . This metric is moreover translation invariant, $d(x + z, y + z) = d(x, y)$, and scales as $d(\alpha x, \alpha y) = |\alpha| d(x, y)$. Hence not every metric comes from a norm: the discrete metric on $\mathbb{R}$ ($d(x, y) = 1$ for $x \neq y$) satisfies $d(2, 0) = d(1, 0) = 1$ and does not scale.

Convergence, Cauchy sequences, open and closed sets in a normed space always refer to the metric $d(x, y) = \|x - y\|$.

PROPOSITION 1.3

Let $(V, \|\cdot\|)$ be a normed space.

- (i) For all $x, y \in V$, $|\|x\| - \|y\|| \leq \|x - y\|$; in particular the norm is continuous on V .
- (ii) If $x_n \rightarrow x$, $y_n \rightarrow y$ and $\alpha_n \rightarrow \alpha$ ($\alpha_n \in \mathbb{K}$), then $x_n + y_n \rightarrow x + y$ and $\alpha_n x_n \rightarrow \alpha x$.

Proof. (i) By the triangle inequality $\|x\| \leq \|x - y\| + \|y\|$; likewise $\|y\| \leq \|y - x\| + \|x\|$, and $\|y - x\| = \|x - y\|$.

(ii) $\|(x_n + y_n) - (x + y)\| \leq \|x_n - x\| + \|y_n - y\| \rightarrow 0$. Moreover

$$\|\alpha_n x_n - \alpha x\| \leq |\alpha_n| \|x_n - x\| + |\alpha_n - \alpha| \|x\|,$$

and $(|\alpha_n|)$ is bounded, so the right-hand side tends to 0. $\square$

1.2 Banach spaces and the series criterion

DEFINITION 1.4 *Banach space*

A sequence (x_n) in a normed space V is a **Cauchy sequence** if for every $\varepsilon > 0$ there is N such that $\|x_m - x_n\| < \varepsilon$ for all $m, n \geq N$. V is **complete** if every Cauchy sequence in V converges to an element of V . A complete normed space is called a **Banach space**.

Series make sense in a normed space. The series $\sum_{k=1}^{\infty} x_k$ **converges** if the partial sums $S_n = \sum_{k=1}^n x_k$ converge in V , and it **converges absolutely** if $\sum_{k=1}^{\infty} \|x_k\| < \infty$. In $\mathbb{R}$ absolute convergence implies convergence, which is one form of the completeness of $\mathbb{R}$. The next theorem shows that, in a general normed space, this property characterises completeness exactly; it will be our tool for proving that L^p is complete.

THEOREM 1.5 *Series criterion for completeness*

A normed space V is complete if and only if every absolutely convergent series in V converges in V .

Proof. Necessity: let V be complete and $\sum \|x_k\| < \infty$. For $m > n$,

$$\|S_m - S_n\| = \left\| \sum_{k=n+1}^m x_k \right\| \leq \sum_{k=n+1}^m \|x_k\| \xrightarrow{n \rightarrow \infty} 0,$$

so (S_n) is a Cauchy sequence and therefore converges.

Sufficiency: let (x_n) be a Cauchy sequence. Choose $n_1 < n_2 < \dots$ such that $\|x_m - x_n\| < 2^{-k}$ whenever $m, n \geq n_k$. Put $y_1 = x_{n_1}$ and $y_{k+1} = x_{n_{k+1}} - x_{n_k}$; then $\sum_k \|y_k\| \leq \|y_1\| + \sum_k 2^{-k} < \infty$. By hypothesis $\sum y_k$ converges to some $x \in V$, and its k -th partial sum is exactly x_{n_k} , so $x_{n_k} \rightarrow x$. Finally, given $\varepsilon > 0$ choose N with $\|x_m - x_n\| < \varepsilon$ for $m, n \geq N$, and then k with $n_k \geq N$ and $\|x_{n_k} - x\| < \varepsilon$. For $n \geq N$,

$$\|x_n - x\| \leq \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < 2\varepsilon.$$

Hence $x_n \rightarrow x$. $\square$

1.3 Finite-dimensional examples: p -norms on $\mathbb{K}^n$

EXAMPLE 1.6 p -norms on $\mathbb{K}^n$

For $x = (x_1, \dots, x_n) \in \mathbb{K}^n$ and $1 \leq p < \infty$ put

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

Definiteness and homogeneity are obvious; for $p = 1$ and $p = \infty$ the triangle inequality follows directly from the triangle inequality in $\mathbb{K}$; for $1 < p < \infty$ it is the discrete Minkowski inequality, proved in Section 3 (Corollary 3.11). The induced distance $d_p(x, y) = \|x - y\|_p$ is often called the **Minkowski distance**; $p = 1, 2, \infty$ give the “Manhattan”, Euclidean and Chebyshev distances. From

$$\|x\|_\infty \leq \|x\|_p \leq n^{1/p} \|x\|_\infty$$

we get $\lim_{p \rightarrow \infty} \|x\|_p = \|x\|_\infty$, which explains the notation $\|\cdot\|_\infty$. Figure 1 shows the unit balls $B_p = \{x : \|x\|_p \leq 1\}$ in $\mathbb{R}^2$.

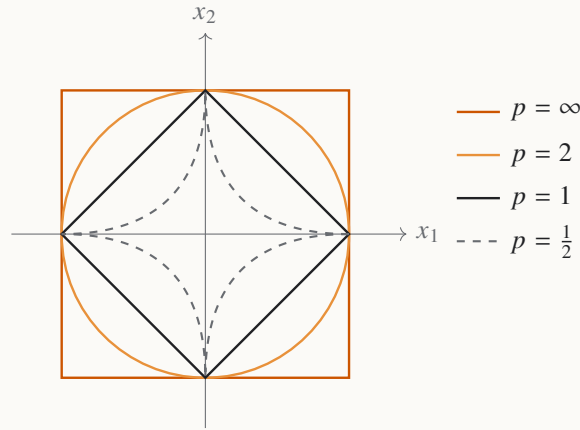


Figure 1. Boundaries of $\{\|x\|_p \leq 1\}$ in $\mathbb{R}^2$. For $p \geq 1$ the unit ball is convex; for $p = \frac{1}{2}$ it is not.

EXAMPLE 1.7 Not a norm for $0 < p < 1$

For $0 < p < 1$ the same formula still defines $\|x\|_p$, and definiteness and homogeneity still hold, but for $n \geq 2$ the triangle inequality fails: with $e_1 = (1, 0, \dots, 0)$ and $e_2 = (0, 1, 0, \dots, 0)$,

$$\|e_1 + e_2\|_p = 2^{1/p} > 2 = \|e_1\|_p + \|e_2\|_p.$$

Geometrically, the unit ball of any norm is convex, since $\|x\|, \|y\| \leq 1$ and $t \in [0, 1]$ imply $\|tx + (1-t)y\| \leq t\|x\| + (1-t)\|y\| \leq 1$; the unit ball for $p = \frac{1}{2}$ in Figure 1 is not convex.

In signal processing one also writes $\|x\|_0 = \#\{i : x_i \neq 0\}$ (the number of nonzero entries). It satisfies the triangle inequality but not homogeneity ($\|2x\|_0 = \|x\|_0$), so it is not a norm either. Note that $\|x\|_0 = \lim_{p \rightarrow 0^+} \|x\|_p^p$: it is the limit of $\|x\|_p^p$, not of $\|x\|_p$.

In finite dimensions all norms are topologically the same.

DEFINITION 1.8 *Equivalent norms*

Two norms $\|\cdot\|$ and $\|\cdot\|'$ on a vector space V are **equivalent** if there are constants $0 < c \leq C$ such that $c\|x\| \leq \|x\|' \leq C\|x\|$ for all $x \in V$.

Equivalent norms have the same convergent sequences, the same Cauchy sequences and the same open sets.

THEOREM 1.9

Any two norms on a finite-dimensional vector space are equivalent.

Proof. Choose a basis $e_1, \dots, e_n$ and identify V with $\mathbb{K}^n$. It suffices to show that an arbitrary norm $\|\cdot\|$ is equivalent to $\|\cdot\|_1$. On the one hand,

$$\|x\| = \left\| \sum_i x_i e_i \right\| \leq \sum_i |x_i| \|e_i\| \leq M \|x\|_1, \quad M = \max_i \|e_i\|.$$

Hence $|\|x\| - \|y\|| \leq M\|x - y\|_1$, i.e. $\|\cdot\|$ is continuous with respect to $\|\cdot\|_1$. On the other hand, regarding $\mathbb{K}^n$ as $\mathbb{R}^n$ or $\mathbb{R}^{2n}$, the Cauchy–Schwarz inequality gives $\|x\|_1 \leq \sqrt{n} \|x\|_2$, and clearly $\|x\|_2 \leq \|x\|_1$; so $\|\cdot\|_1$ and the Euclidean norm have the same bounded sets and the same closed sets. Therefore the unit sphere $S = \{x : \|x\|_1 = 1\}$ is closed and bounded in the Euclidean sense, hence compact by the Heine–Borel theorem (Article I). The continuous function $\|\cdot\|$ attains a minimum c on S , and $c > 0$ because $0 \notin S$. For $x \neq 0$,

$$\|x\| = \|x\|_1 \left\| \frac{x}{\|x\|_1} \right\| \geq c \|x\|_1.$$

□

COROLLARY 1.10

Every finite-dimensional normed space is complete; every finite-dimensional subspace of a normed space is closed.

Proof. Let $(x^{(k)})$ be a Cauchy sequence in a finite-dimensional normed space. By Theorem 1.9 it is also Cauchy for $\|\cdot\|_1$, so each coordinate sequence $(x_i^{(k)})_k$ is Cauchy in $\mathbb{K}$ and converges, by completeness of $\mathbb{K}$, to some x_i . Thus $\|x^{(k)} - x\|_1 \rightarrow 0$, and by equivalence $\|x^{(k)} - x\| \rightarrow 0$. The second statement follows because a complete subset of a metric space is closed. □

REMARK 1.11

The common phrase “an n -dimensional normed space is isomorphic to $\mathbb{K}^n$ ” should be read precisely: an n -dimensional normed space over $\mathbb{K}$ is linearly homeomorphic to $(\mathbb{K}^n, \|\cdot\|_2)$, but in general *not* isometrically isomorphic. For example, there is no linear isometry between $(\mathbb{R}^2, \|\cdot\|_\infty)$ and $(\mathbb{R}^2, \|\cdot\|_2)$: a linear map sends a disc onto an elliptical disc, never onto a square.

1.4 Infinite-dimensional examples

EXAMPLE 1.12 $C[a, b]$ with the supremum norm

Let $C[a, b]$ be the space of continuous $\mathbb{K}$ -valued functions on $[a, b]$ and $\|f\|_u = \max_{x \in [a, b]} |f(x)|$ (the **uniform** or supremum norm). Then $(C[a, b], \|\cdot\|_u)$ is a Banach space.

Indeed, let (f_n) be a Cauchy sequence. For each x , $|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_u$, so $(f_n(x))$ is a Cauchy sequence in $\mathbb{K}$; call its limit $f(x)$. Given $\varepsilon > 0$, choose N with $\|f_n - f_m\|_u < \varepsilon$ for $m, n \geq N$; letting $m \rightarrow \infty$ in $|f_n(x) - f_m(x)| < \varepsilon$ gives $\sup_x |f_n(x) - f(x)| \leq \varepsilon$ for $n \geq N$, i.e. $f_n \rightarrow f$ uniformly. A uniform limit of continuous functions is continuous: for $x_0 \in [a, b]$ choose $\delta > 0$ with $|f_N(x) - f_N(x_0)| < \varepsilon$ whenever $|x - x_0| < \delta$; then

$$|f(x) - f(x_0)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| < 3\varepsilon.$$

Hence $f \in C[a, b]$ and $\|f_n - f\|_u \rightarrow 0$.

EXAMPLE 1.13 An incomplete norm on the same space

On $C[0, 2]$ put $\|f\|_1 = \int_0^2 |f(x)| dx$. This is a norm: if f is continuous and $|f(x_0)| > 0$, then $|f| > |f(x_0)|/2$ on a neighbourhood of x_0 , so $\|f\|_1 > 0$. But $(C[0, 2], \|\cdot\|_1)$ is not complete. Let

$$f_n(x) = \begin{cases} 0, & 0 \leq x \leq 1, \\ n(x-1), & 1 < x < 1 + \frac{1}{n}, \\ 1, & 1 + \frac{1}{n} \leq x \leq 2. \end{cases}$$

For $m > n$, f_m and f_n differ only on $[1, 1 + \frac{1}{n}]$, and there by at most 1; so $\|f_m - f_n\|_1 \leq \frac{1}{n}$ and (f_n) is Cauchy. If some continuous f satisfied $\|f_n - f\|_1 \rightarrow 0$, then $\int_0^1 |f| \leq \|f - f_n\|_1 \rightarrow 0$, so $f = 0$ on $[0, 1]$; and for every $\delta \in (0, 1)$ and $n > 1/\delta$, $\int_{1+\delta}^2 |f - 1| \leq \|f - f_n\|_1 \rightarrow 0$, so $f = 1$ on $[1 + \delta, 2]$ and hence on $(1, 2]$. This contradicts continuity of f at 1.

Thus the same vector space can be complete for one norm and incomplete for another; in particular $\|\cdot\|_u$ and $\|\cdot\|_1$ are not equivalent on $C[0, 2]$, in contrast with Theorem 1.9. We shall see in Section 7 that the completion of this space is precisely $L^1[0, 2]$.

EXAMPLE 1.14 Sequence spaces

For $1 \leq p < \infty$ let ℓ^p be the space of $\mathbb{K}$ -valued sequences $x = (x_n)_{n \geq 1}$ with $\sum_n |x_n|^p < \infty$, with $\|x\|_p = (\sum_n |x_n|^p)^{1/p}$; let ℓ^∞ be the space of bounded sequences with $\|x\|_\infty = \sup_n |x_n|$. Example 2.9 will show that these are the L^p spaces of counting measure, so that everything proved below for L^p (Minkowski's inequality, completeness, ...) applies to ℓ^p .

1.5 Bounded linear maps

DEFINITION 1.15

Let V, W be normed spaces over $\mathbb{K}$. A linear map $T : V \rightarrow W$ is **bounded** if there is a constant $C \geq 0$ with $\|Tx\|_W \leq C\|x\|_V$ for all $x \in V$. The smallest such constant,

$$\|T\| = \sup_{\|x\|_V \leq 1} \|Tx\|_W,$$

is the **operator norm** of T . For $W = \mathbb{K}$ a bounded linear map is called a bounded linear functional; the space of all bounded linear functionals on V is denoted V^* and called the **dual space** of V . With the operator norm it is a Banach space (the proof is similar to Example 1.12 and uses only the completeness of $\mathbb{K}$).

PROPOSITION 1.16

For a linear map $T : V \rightarrow W$ the following are equivalent: (a) T is bounded; (b) T is continuous; (c) T is continuous at 0.

Proof. (a) $\Rightarrow$ (b): $\|Tx - Ty\|_W = \|T(x - y)\|_W \leq C\|x - y\|_V$, so T is Lipschitz. (b) $\Rightarrow$ (c) is trivial. (c) $\Rightarrow$ (a): there is $\delta > 0$ with $\|Tx\|_W \leq 1$ whenever $\|x\|_V \leq \delta$. For $x \neq 0$,

$$\|Tx\|_W = \frac{\|x\|_V}{\delta} \left\| T\left(\frac{\delta x}{\|x\|_V}\right) \right\|_W \leq \frac{1}{\delta} \|x\|_V.$$

□

2 L^p Spaces

2.1 The p -norm and the essential supremum

DEFINITION 2.1

Let f be a measurable function on X with values in $\mathbb{K}$ or $[-\infty, \infty]$. For $0 < p < \infty$ put

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p} \in [0, \infty].$$

For $p = \infty$ put

$$\|f\|_\infty = \operatorname{ess\,sup}_X |f| = \inf \{ M \geq 0 : \mu(\{x : |f(x)| > M\}) = 0 \} \in [0, \infty],$$

with $\inf \emptyset = \infty$. $\|f\|_\infty$ is called the **essential supremum** of $|f|$; it is the smallest a.e. upper bound of $|f|$.

REMARK 2.2

The condition “ $\mu(\{|f| > M\}) = 0$ ” (i.e. $|f| \leq M$ a.e.) cannot be dropped: requiring $|f| \leq M$ everywhere yields the ordinary supremum $\sup |f|$, which is changed by altering f on a null set. For example $f = \mathbf{1}_{\mathbb{Q}}$ on $\mathbb{R}$ has $\sup |f| = 1$ but $\|f\|_{\infty} = 0$.

LEMMA 2.3

For every measurable f , $|f| \leq \|f\|_{\infty}$ a.e.; that is, the infimum in the definition is attained.

Proof. If $\|f\|_{\infty} = \infty$ there is nothing to prove. Otherwise

$$\{|f| > \|f\|_{\infty}\} = \bigcup_{k=1}^{\infty} \left\{ |f| > \|f\|_{\infty} + \frac{1}{k} \right\},$$

each set on the right is null by the definition of the infimum, and a countable union of null sets is null. $\square$

LEMMA 2.4

Let $0 < p \leq \infty$ and let f, g be measurable.

- (i) $\|f\|_p = 0$ if and only if $f = 0$ a.e.;
- (ii) if $f = g$ a.e., then $\|f\|_p = \|g\|_p$;
- (iii) if f is $\mathbb{K}$ -valued and $\alpha \in \mathbb{K}$, then $\|\alpha f\|_p = |\alpha| \|f\|_p$.

Proof. (i) For $p < \infty$: by Article II, the integral of the nonnegative measurable function $|f|^p$ vanishes if and only if $|f|^p = 0$ a.e. For $p = \infty$: if $\|f\|_{\infty} = 0$, Lemma 2.3 gives $|f| \leq 0$ a.e.; the converse is clear. (ii) For $p < \infty$, $|f|^p = |g|^p$ a.e., so the integrals coincide; for $p = \infty$, $\{|f| > M\}$ and $\{|g| > M\}$ differ by a null set. (iii) Follows from homogeneity of the integral ($p < \infty$) or from $\{|\alpha f| > |\alpha| M\} = \{|f| > M\}$ ($p = \infty, \alpha \neq 0$). $\square$

2.2 Definition of L^p **DEFINITION 2.5**

For $0 < p \leq \infty$ let

$$\mathcal{L}^p(X, \mu) = \{f : X \rightarrow \mathbb{K} \mid f \text{ measurable, } \|f\|_p < \infty\}.$$

PROPOSITION 2.6

$\mathcal{L}^p(X, \mu)$ is a vector space over $\mathbb{K}$ under pointwise addition and scalar multiplication. More precisely, for $a, b \in \mathbb{K}$:

$$|a + b|^p \leq 2^{p-1}(|a|^p + |b|^p) \quad (1 \leq p < \infty), \quad |a + b|^p \leq |a|^p + |b|^p \quad (0 < p \leq 1).$$

Proof. Closure under scalar multiplication is Lemma 2.4(iii). For $1 \leq p < \infty$ the function $t \mapsto t^p$ is convex on $[0, \infty)$, so

$$\left(\frac{|a| + |b|}{2}\right)^p \leq \frac{|a|^p + |b|^p}{2},$$

which together with $|a + b| \leq |a| + |b|$ gives the first inequality. For $0 < p \leq 1$ and $s, t \geq 0$ with $s + t > 0$, since $\frac{s}{s+t}, \frac{t}{s+t} \in [0, 1]$ and $p \leq 1$,

$$1 = \frac{s}{s+t} + \frac{t}{s+t} \leq \left(\frac{s}{s+t}\right)^p + \left(\frac{t}{s+t}\right)^p,$$

i.e. $(s+t)^p \leq s^p + t^p$; taking $s = |a|, t = |b|$ gives the second inequality. Applying these inequalities with $a = f(x), b = g(x)$ and integrating shows that $f + g \in \mathcal{L}^p$ whenever $f, g \in \mathcal{L}^p$ ($p < \infty$). For $p = \infty$, Lemma 2.3 gives $|f + g| \leq \|f\|_\infty + \|g\|_\infty$ a.e. $\square$

However, $\|\cdot\|_p$ is in general not definite on $\mathcal{L}^p$: for instance $f = \mathbf{1}_{\{0\}}$ on $(\mathbb{R}, m)$ has $\|f\|_p = 0$ although $f \neq 0$. The remedy is not to distinguish between functions that agree almost everywhere. The relation “ $f \sim g \iff f = g$ a.e.” is an equivalence relation on $\mathcal{L}^p$ (transitivity holds because the union of two null sets is null), and $\mathcal{N} = \{f : f = 0 \text{ a.e.}\}$ is a linear subspace of $\mathcal{L}^p$.

DEFINITION 2.7 L^p spaces

For $0 < p \leq \infty$ define the quotient space

$$L^p(X, \mu) = \mathcal{L}^p(X, \mu) / \mathcal{N},$$

whose elements are the equivalence classes $[f] = \{g \in \mathcal{L}^p : g = f \text{ a.e.}\}$, with operations $[f] + [g] = [f + g]$, $\alpha[f] = [\alpha f]$, and put $\|[f]\|_p = \|f\|_p$.

By Lemma 2.4(ii), $\|[f]\|_p$ does not depend on the choice of representative, and by Lemma 2.4(i), $\|[f]\|_p = 0$ if and only if $[f] = [0]$. Consequently, as soon as the triangle inequality is proved for $1 \leq p \leq \infty$ (Section 3), $L^p(X, \mu)$ is a normed space.

REMARK 2.8 Conventions

- (i) As usual we identify the class $[f]$ with a representative f and write $f \in L^p$; relations such as $f \geq 0, f \leq g, f = g$ are understood to hold almost everywhere. One must remember, however, that elements of L^p are not functions: when points are null sets (as in $(\mathbb{R}^n, m)$), “the value at 0 of $f \in L^p(\mathbb{R})$ ” is meaningless.
- (ii) If f takes values in $[-\infty, \infty]$ and $\|f\|_p < \infty$ ($p < \infty$), then f is finite a.e. (otherwise $|f|^p = \infty$ on a set of positive measure), so it agrees a.e. with a finite-valued function; allowing extended real-valued representatives therefore changes nothing.
- (iii) L^p is defined for every measure space, not only for Lebesgue measure. When $X = \mathbb{R}^n$, or a measurable subset E , carries Lebesgue measure we write $L^p(\mathbb{R}^n), L^p(E), L^p[a, b]$, etc.; when no confusion can arise we simply write $L^p(X)$ or L^p .

2.3 Two basic examples

EXAMPLE 2.9 Counting measure and ℓ^p

Let $X = \mathbb{N}$, $\mathcal{M} = \mathcal{P}(\mathbb{N})$ and μ counting measure ($\mu(A) = \#A$). Every function $f : \mathbb{N} \rightarrow \mathbb{K}$ is measurable. For nonnegative f , the simple functions $f\mathbf{1}_{\{1, \dots, N\}}$ increase to f , so by the monotone convergence theorem

$$\int_{\mathbb{N}} f \, d\mu = \lim_{N \rightarrow \infty} \sum_{n=1}^N f(n) = \sum_{n=1}^{\infty} f(n).$$

Since the only null set is $\emptyset$, each equivalence class contains a single function, and $\|f\|_{\infty} = \sup_n |f(n)|$. Hence $L^p(\mathbb{N}, \mu) = \ell^p$ ($0 < p \leq \infty$) with the norms of Example 1.14. Likewise, counting measure on $X = \{1, \dots, n\}$ gives $(\mathbb{K}^n, \|\cdot\|_p)$ of Example 1.6.

EXAMPLE 2.10 Power functions

Let $\alpha > 0$ and $0 < p < \infty$. Then

$$x^{-\alpha} \in L^p(0, 1) \iff \alpha p < 1, \quad x^{-\alpha} \in L^p(1, \infty) \iff \alpha p > 1.$$

Indeed, $x^{-\alpha p} \mathbf{1}_{[\varepsilon, 1]}$ increases to $x^{-\alpha p} \mathbf{1}_{(0, 1)}$ as $\varepsilon \downarrow 0$; by the monotone convergence theorem and the equality of the Lebesgue and Riemann integrals of a continuous function on a closed interval (Article II),

$$\int_{(0, 1)} x^{-\alpha p} \, dx = \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^1 x^{-\alpha p} \, dx = \begin{cases} \frac{1}{1 - \alpha p}, & \alpha p < 1, \\ \infty, & \alpha p \geq 1. \end{cases}$$

The case of $(1, \infty)$ is analogous. Thus $x^{-\alpha} \mathbf{1}_{(0, 1)}$ belongs only to L^p with small exponents (a local singularity penalises large p), whereas $x^{-\alpha} \mathbf{1}_{(1, \infty)}$ belongs only to L^p with large exponents (slow decay at infinity penalises small p). This contrast will recur in Section 6.

3 The Inequalities of Young, Hölder and Minkowski

3.1 Young's inequality

LEMMA 3.1 Young's inequality

Let $1 < p < \infty$, let q be the conjugate exponent and $a, b \geq 0$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

with equality if and only if $a^p = b^q$.

Proof. If $a = 0$ or $b = 0$, the left side is 0 and the right side is nonnegative, and the right side

vanishes if and only if $a = b = 0$, i.e. if and only if $a^p = b^q$. Let $a, b > 0$. Since $(e^t)'' = e^t > 0$, the exponential function is strictly convex: for $\lambda \in (0, 1)$,

$$e^{\lambda s + (1-\lambda)t} \leq \lambda e^s + (1-\lambda)e^t,$$

with equality if and only if $s = t$. Take $\lambda = \frac{1}{p}$ (so $1 - \lambda = \frac{1}{q}$), $s = \log a^p$, $t = \log b^q$. The left side is $e^{\log a + \log b} = ab$ and the right side is $\frac{a^p}{p} + \frac{b^q}{q}$; equality holds if and only if $a^p = b^q$. $\square$

REMARK 3.2

Young's inequality has an intuitive geometric interpretation (Figure 2): the curve $y = x^{p-1}$ and its inverse $x = y^{q-1}$ (note $(p-1)(q-1) = 1$) bound two regions which together cover the rectangle $[0, a] \times [0, b]$; their areas are $\int_0^a x^{p-1} dx = \frac{a^p}{p}$ and $\int_0^b y^{q-1} dy = \frac{b^q}{q}$. The two regions fit together exactly into the rectangle if and only if $b = a^{p-1}$, i.e. $a^p = b^q$.

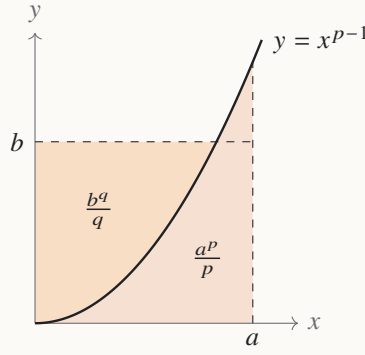


Figure 2. Geometric meaning of Young's inequality ($p = 3$, $q = \frac{3}{2}$). The two shaded regions cover the rectangle $[0, a] \times [0, b]$.

3.2 Hölder's inequality

Recall the Cauchy–Schwarz inequality $|\langle u, v \rangle| \leq \|u\| \|v\|$ in an inner product space V . Hölder's inequality is its generalisation to the L^p setting.

THEOREM 3.3 Hölder's inequality

Let $1 \leq p \leq \infty$, let q be the conjugate exponent and let f, g be measurable $\mathbb{K}$ -valued functions on X . Then

$$\|fg\|_1 = \int_X |fg| d\mu \leq \|f\|_p \|g\|_q$$

(both sides in $[0, \infty]$). In particular, if $f \in L^p$ and $g \in L^q$, then $fg \in L^1$ and $|\int_X fg d\mu| \leq \|f\|_p \|g\|_q$.

Proof. Case $p = 1$, $q = \infty$: by Lemma 2.3, $|fg| \leq \|g\|_\infty |f|$ a.e. (with $0 \cdot \infty = 0$); integrate. The case $p = \infty$ is symmetric.

Case $1 < p < \infty$. If $\|f\|_p = 0$ or $\|g\|_q = 0$, then by Lemma 2.4 $f = 0$ a.e. or $g = 0$ a.e., so $fg = 0$ a.e. and both sides vanish. If neither is 0 and one of them is ∞ , the right side is ∞ and there is

nothing to prove. So assume $A = \|f\|_p$ and $B = \|g\|_q$ lie in $(0, \infty)$. For each x apply Young's inequality with $a = |f(x)|/A$, $b = |g(x)|/B$:

$$\frac{|f(x)g(x)|}{AB} \leq \frac{1}{p} \frac{|f(x)|^p}{A^p} + \frac{1}{q} \frac{|g(x)|^q}{B^q}.$$

Integrating, the right side becomes $\frac{1}{p} \cdot 1 + \frac{1}{q} \cdot 1 = 1$, so $\|fg\|_1 \leq AB$. The last assertion follows from $|\int h| \leq \int |h|$ (see the Conventions). $\square$

REMARK 3.4

The logical order of the proof is: normalise f and g by their norms, apply Young's inequality pointwise to the normalised functions, then integrate. Normalisation requires $0 < \|f\|_p, \|g\|_q < \infty$, so the cases of norm 0 or ∞ must be treated separately. Note also that the two functions are measured with the exponents p and q respectively: the right-hand side is $(\int |f|^p)^{1/p} (\int |g|^q)^{1/q}$, with $1 \leq p, q \leq \infty$.

COROLLARY 3.5 *Cauchy–Schwarz inequality*

If $f, g \in L^2(X, \mu)$, then $|\int_X f \bar{g} \, d\mu| \leq \|f\|_2 \|g\|_2$. Hence $\langle f, g \rangle = \int_X f \bar{g} \, d\mu$ is an inner product on L^2 with $\langle f, f \rangle = \|f\|_2^2$.

This holds for every measure space, not only for $L^2(S^1)$. Together with completeness (Section 4), $L^2(X, \mu)$ is a Hilbert space. For counting measure we obtain the discrete form

$$\sum_n |x_n y_n| \leq \left(\sum_n |x_n|^p \right)^{1/p} \left(\sum_n |y_n|^q \right)^{1/q}.$$

COROLLARY 3.6 *Generalised Hölder inequality*

Let $0 < p, q, r \leq \infty$ with $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$, and let f, g be measurable. Then $\|fg\|_r \leq \|f\|_p \|g\|_q$.

Proof. If $r = \infty$ then $p = q = \infty$, and $|fg| \leq \|f\|_\infty \|g\|_\infty$ a.e. by Lemma 2.3. Let $r < \infty$. If $p = \infty$, then $q = r$ and $|fg|^r \leq \|f\|_\infty^r |g|^r$ a.e.; integrate. The case $q = \infty$ is the same. If $p, q < \infty$, then $\frac{p}{r}, \frac{q}{r} > 1$ and $\frac{r}{p} + \frac{r}{q} = 1$; applying Theorem 3.3 to $|f|^r$ and $|g|^r$ with exponents $\frac{p}{r}, \frac{q}{r}$ gives

$$\int |f|^r |g|^r \, d\mu \leq \left(\int |f|^p \, d\mu \right)^{r/p} \left(\int |g|^q \, d\mu \right)^{r/q},$$

and we take r -th roots. $\square$

By induction, if $\frac{1}{r} = \sum_{i=1}^k \frac{1}{p_i}$, then $\|f_1 \cdots f_k\|_r \leq \prod_i \|f_i\|_{p_i}$.

3.3 Equality in Hölder's inequality

THEOREM 3.7

- (i) Let $1 < p < \infty$, $f \in L^p$, $g \in L^q$. Then $\|fg\|_1 = \|f\|_p \|g\|_q$ if and only if there are constants $\alpha, \beta \geq 0$, not both zero, with $\alpha|f|^p = \beta|g|^q$ a.e. When $\|f\|_p, \|g\|_q > 0$ this is equivalent to

$$\frac{|f|^p}{\|f\|_p^p} = \frac{|g|^q}{\|g\|_q^q} \quad \text{a.e.}$$

- (ii) Let $f \in L^1$, $g \in L^\infty$. Then $\|fg\|_1 = \|f\|_1 \|g\|_\infty$ if and only if $|g| = \|g\|_\infty$ a.e. on the set $\{f \neq 0\}$.

Proof. (i) If $\|f\|_p = 0$, both sides of the equality are 0, and $\alpha = 1, \beta = 0$ satisfy the condition; conversely, if the condition holds with $\beta = 0$, then $\alpha > 0$, $f = 0$ a.e., and equality holds. The cases $\|g\|_q = 0$ or $\alpha = 0$ are symmetric. Assume now $A = \|f\|_p > 0$ and $B = \|g\|_q > 0$, and let $F = |f|/A$, $G = |g|/B$,

$$H = \frac{F^p}{p} + \frac{G^q}{q} - FG.$$

By Young's inequality $H \geq 0$ a.e., and by the proof of Theorem 3.3, $\int H d\mu = 1 - \frac{\|fg\|_1}{AB}$. Hence equality holds $\iff \int H = 0 \iff H = 0$ a.e. (Article II) $\iff F^p = G^q$ a.e. (the equality case of Young's inequality), i.e. $|f|^p/A^p = |g|^q/B^q$ a.e.; this is the condition with $\alpha = A^{-p}$, $\beta = B^{-q}$. Conversely, if $\alpha|f|^p = \beta|g|^q$ a.e. with α, β not both zero, then since $A, B > 0$ both α and β must be positive (e.g. $\alpha = 0$ would force $g = 0$ a.e.); integrating gives $\alpha A^p = \beta B^q$, whence $|f|^p/A^p = |g|^q/B^q$ a.e.

- (ii) By Hölder's inequality $\|fg\|_1 < \infty$, and

$$\|f\|_1 \|g\|_\infty - \|fg\|_1 = \int_X |f|(\|g\|_\infty - |g|) d\mu,$$

where the integrand is nonnegative a.e. (Lemma 2.3). So the difference vanishes if and only if $|f|(\|g\|_\infty - |g|) = 0$ a.e., i.e. $|g| = \|g\|_\infty$ a.e. on $\{f \neq 0\}$. $\square$

EXAMPLE 3.8

For $p = 1, q = \infty$ the equality condition is sometimes misstated as “ f and g have the same support a.e.”. This is neither sufficient nor necessary. On $[0, 1]$ take $f \equiv 1$ and $g(x) = x$: the sets $\{f \neq 0\}$ and $\{g \neq 0\}$ differ by a null set, yet $\|fg\|_1 = \frac{1}{2} < 1 = \|f\|_1 \|g\|_\infty$. Conversely, take $f = \mathbf{1}_{[0, 1/2]}$ and $g \equiv 1$: the supports differ, yet $\|fg\|_1 = \frac{1}{2} = \|f\|_1 \|g\|_\infty$. For $1 < p < \infty$, the condition “ $|f|^p = \lambda|g|^q$ for some $\lambda \geq 0$ ” misses the case $g = 0, f \neq 0$ (where equality $0 = 0$ holds trivially); this is why the theorem uses two coefficients α, β , not both zero.

3.4 Minkowski's inequality

THEOREM 3.9 *Minkowski's inequality*

Let $1 \leq p \leq \infty$ and $f, g \in L^p(X, \mu)$. Then $f + g \in L^p$ and

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Consequently, for $1 \leq p \leq \infty$, $\|\cdot\|_p$ is a seminorm on $\mathcal{L}^p$ and $L^p(X, \mu)$ is a normed space.

Proof. $p = 1$: integrate $|f + g| \leq |f| + |g|$. $p = \infty$: by Lemma 2.3, outside a null set $|f + g| \leq |f| + |g| \leq \|f\|_\infty + \|g\|_\infty$.

Let $1 < p < \infty$. By Proposition 2.6, $f + g \in L^p$, i.e. $\|f + g\|_p < \infty$; if $\|f + g\|_p = 0$ there is nothing to prove, so assume $0 < \|f + g\|_p < \infty$. Since $(p - 1)q = p$,

$$\|f + g\|_p^{p-1} = \left(\int |f + g|^p d\mu \right)^{1/q} = \|f + g\|_p^{p/q} = \|f + g\|_p^{p-1} < \infty.$$

Using $|f + g| \leq |f| + |g|$ and Hölder's inequality,

$$\begin{aligned} \|f + g\|_p^p &= \int |f + g| |f + g|^{p-1} d\mu \leq \int |f| |f + g|^{p-1} d\mu + \int |g| |f + g|^{p-1} d\mu \\ &\leq (\|f\|_p + \|g\|_p) \|f + g\|_p^{p-1}. \end{aligned}$$

Dividing by $\|f + g\|_p^{p-1} \in (0, \infty)$ gives the result. $\square$

REMARK 3.10

The preliminary step “ $\|f + g\|_p < \infty$ ” is essential: without it the final step might be a division by ∞ . The proof above is valid on every measure space; finite sums and series are merely the special case of counting measure.

COROLLARY 3.11 *Discrete Minkowski inequality*

For $1 \leq p < \infty$ and $\mathbb{K}$ -valued sequences (finite or infinite) $(x_k), (y_k)$,

$$\left(\sum_k |x_k + y_k|^p \right)^{1/p} \leq \left(\sum_k |x_k|^p \right)^{1/p} + \left(\sum_k |y_k|^p \right)^{1/p}.$$

In particular $(\mathbb{K}^n, \|\cdot\|_p)$ and ℓ^p ($1 \leq p \leq \infty$) are normed spaces.

Proof. Apply Theorem 3.9 to counting measure on $\{1, \dots, n\}$ or $\mathbb{N}$ (Example 2.9); if the right side is ∞ there is nothing to prove. $\square$

The “integral form” of Minkowski's inequality often quoted, $(\int_a^b |f + g|^p)^{1/p} \leq (\int_a^b |f|^p)^{1/p} + (\int_a^b |g|^p)^{1/p}$, is just Theorem 3.9 for $X = [a, b]$; it requires only $f, g \in L^p[a, b]$, $1 \leq p < \infty$.

THEOREM 3.12 *Equality in Minkowski's inequality*

Let $1 < p < \infty$ and $f, g \in L^p$. Then $\|f + g\|_p = \|f\|_p + \|g\|_p$ if and only if there is $\lambda \geq 0$ with $f = \lambda g$ a.e. or $g = \lambda f$ a.e.

Proof. Sufficiency: if $g = \lambda f$, then $\|f + g\|_p = (1 + \lambda)\|f\|_p = \|f\|_p + \|g\|_p$.

Necessity: if $f = 0$ a.e., then $f = 0 \cdot g$; similarly if $g = 0$ a.e. Let $\|f\|_p, \|g\|_p > 0$, so that $\|f + g\|_p = \|f\|_p + \|g\|_p > 0$. Put $h = |f + g|^{p-1}$; then $\|h\|_q = \|f + g\|_p^{p-1} > 0$. In the proof of Theorem 3.9,

$$\|f + g\|_p^p \leq \int (|f| + |g|)h \, d\mu \leq (\|f\|_p + \|g\|_p)\|h\|_q = \|f + g\|_p^p,$$

so both inequalities are equalities.

Equality in the first gives $\int (|f| + |g| - |f + g|)h \, d\mu = 0$ with a nonnegative integrand, so $|f + g| = |f| + |g|$ a.e. on $S = \{f + g \neq 0\}$. The second is the sum of the two Hölder inequalities $\int |f|h \leq \|f\|_p\|h\|_q$ and $\int |g|h \leq \|g\|_p\|h\|_q$, so both are equalities. By Theorem 3.7(i),

$$\frac{|f|^p}{\|f\|_p^p} = \frac{h^q}{\|h\|_q^q} = \frac{|f + g|^p}{\|f + g\|_p^p} \quad \text{a.e.,}$$

i.e. $|f| = s|f + g|$ a.e. with $s = \|f\|_p/\|f + g\|_p > 0$; likewise $|g| = t|f + g|$ a.e. with $t = \|g\|_p/\|f + g\|_p > 0$. In particular $f = g = 0$ a.e. on $X \setminus S$.

On S , for complex numbers $z = f(x)$, $w = g(x)$, the identity $|z + w|^2 = |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w})$ shows that $|z + w| = |z| + |w|$ is equivalent to $z\bar{w} = |z||w| \geq 0$, i.e. one of z, w is a nonnegative multiple of the other; consequently $z = \frac{|z|}{|z+w|}(z + w)$. Therefore $f = s(f + g)$ and $g = t(f + g)$ a.e. on S , and these identities also hold a.e. on $X \setminus S$. Hence $tf = st(f + g) = sg$ a.e., i.e. $f = \frac{s}{t}g$ a.e. $\square$

REMARK 3.13

For $p = 1$, $\|f + g\|_1 = \|f\|_1 + \|g\|_1$ if and only if $|f + g| = |f| + |g|$ a.e., i.e. $f\bar{g} \geq 0$ a.e.; this does not force f and g to be proportional (any two nonnegative functions give equality). For $p = \infty$ there is no proportionality characterisation either: $f = \mathbf{1}_{[0,1]}$, $g = \mathbf{1}_{[0,2]}$ satisfy $\|f + g\|_\infty = 2 = \|f\|_\infty + \|g\|_\infty$.

3.5 The case $0 < p < 1$ **PROPOSITION 3.14** *Reverse Minkowski inequality*

Let $0 < p < 1$ and let f, g be nonnegative measurable functions. Then $\|f + g\|_p \geq \|f\|_p + \|g\|_p$.

Proof. Since $f + g \geq f$ and $f + g \geq g$, if $\|f\|_p$ or $\|g\|_p$ is ∞ so is the left side, and if one of them is 0 the claim is also clear. Let $a = \|f\|_p$ and $b = \|g\|_p$ lie in $(0, \infty)$, and put $u = f/a$, $v = g/b$, $\lambda = \frac{a}{a+b}$. Then

$$f + g = (a + b)(\lambda u + (1 - \lambda)v).$$

The function $t \mapsto t^p$ is concave on $[0, \infty)$, so $(\lambda u + (1 - \lambda)v)^p \geq \lambda u^p + (1 - \lambda)v^p$. Integrating and using $\int u^p = \int v^p = 1$ gives $\int (f + g)^p d\mu \geq (a + b)^p$. $\square$

EXAMPLE 3.15

If X contains disjoint measurable sets A, B with $0 < \mu(A), \mu(B) < \infty$, then for $0 < p < 1$

$$\|\mathbf{1}_A + \mathbf{1}_B\|_p = (\mu(A) + \mu(B))^{1/p} > \mu(A)^{1/p} + \mu(B)^{1/p} = \|\mathbf{1}_A\|_p + \|\mathbf{1}_B\|_p,$$

the strict inequality coming from $(s + t)^r > s^r + t^r$ for $r = \frac{1}{p} > 1$ and $s, t > 0$. So in this situation $\|\cdot\|_p$ is not a seminorm.

REMARK 3.16

Note that the reverse inequality holds only for *nonnegative* functions and $0 < p < 1$ (for general functions take $g = -f$); for $p = 1$ one has the usual triangle inequality, and for $p = 0$ the formula is meaningless. On the other hand, by Proposition 2.6, for $0 < p < 1$

$$d_p(f, g) = \int_X |f - g|^p d\mu$$

satisfies the triangle inequality and is a translation-invariant metric on L^p , and the method of Theorem 4.1 shows that (L^p, d_p) is complete. In general, however, L^p ($0 < p < 1$) is not normable; for instance, one can show that the only continuous linear functional on $L^p[0, 1]$ is zero. From now on we mainly consider $1 \leq p \leq \infty$.

3.6 The parallelogram law

PROPOSITION 3.17

Suppose X contains disjoint measurable sets A, B with $0 < \mu(A), \mu(B) < \infty$, and let $1 \leq p \leq \infty$. Then the norm of $L^p(X, \mu)$ satisfies the parallelogram law

$$\|f + g\|_p^2 + \|f - g\|_p^2 = 2\|f\|_p^2 + 2\|g\|_p^2 \quad (\forall f, g \in L^p)$$

if and only if $p = 2$.

Proof. For $p = 2$, expand $\|f \pm g\|_2^2 = \|f\|_2^2 \pm 2\operatorname{Re}\langle f, g \rangle + \|g\|_2^2$ and add. Let $p \neq 2$. For $p < \infty$ take $f = \mu(A)^{-1/p} \mathbf{1}_A$, $g = \mu(B)^{-1/p} \mathbf{1}_B$; for $p = \infty$ take $f = \mathbf{1}_A$, $g = \mathbf{1}_B$. Then $\|f\|_p = \|g\|_p = 1$ and the right side equals 4. As A, B are disjoint, $|f \pm g|^p = |f|^p + |g|^p$, so for $p < \infty$ we get $\|f \pm g\|_p = 2^{1/p}$ and the left side is $2 \cdot 2^{2/p} \neq 4$; for $p = \infty$, $\|f \pm g\|_\infty = 1$ and the left side is $2 \neq 4$. $\square$

By the Jordan–von Neumann theorem (a norm comes from an inner product if and only if it satisfies the parallelogram law), under the above hypothesis L^2 is the only L^p space whose norm comes from an inner product.

4 Completeness: the Riesz–Fischer Theorem

THEOREM 4.1 *Riesz–Fischer*

For $1 \leq p < \infty$, $L^p(X, \mu)$ is a Banach space.

Proof. By Theorem 1.5 it suffices to show that every absolutely convergent series in L^p converges. Let $f_k \in L^p$ with $\sum_{k=1}^{\infty} \|f_k\|_p = S < \infty$, and put

$$G_n = \sum_{k=1}^n |f_k|, \quad G = \sum_{k=1}^{\infty} |f_k| : X \rightarrow [0, \infty].$$

By Minkowski's inequality, $\|G_n\|_p \leq \sum_{k=1}^n \|f_k\|_p \leq S$. Since $G_n^p \uparrow G^p$, the monotone convergence theorem gives

$$\int_X G^p d\mu = \lim_{n \rightarrow \infty} \int_X G_n^p d\mu \leq S^p < \infty.$$

Hence $G < \infty$ a.e.: with $E = \{G < \infty\}$ we have $\mu(X \setminus E) = 0$. For $x \in E$ the numerical series $\sum_k f_k(x)$ converges absolutely, hence converges by completeness of $\mathbb{K}$. Define

$$F(x) = \begin{cases} \sum_{k=1}^{\infty} f_k(x), & x \in E, \\ 0, & x \notin E. \end{cases}$$

F is the pointwise limit of the measurable functions $\mathbf{1}_E \sum_{k \leq n} f_k$, hence measurable, and $|F| \leq G$, so $F \in L^p$. Finally, with $S_n = \sum_{k=1}^n f_k$, on E we have

$$|F - S_n| = \left| \sum_{k > n} f_k \right| \leq \sum_{k > n} |f_k| \leq G, \quad |F - S_n| \rightarrow 0.$$

Thus $|F - S_n|^p \rightarrow 0$ a.e. and $|F - S_n|^p \leq G^p \in L^1$, and the dominated convergence theorem gives $\|F - S_n\|_p \rightarrow 0$. So $\sum f_k$ converges to F in L^p . $\square$

REMARK 4.2

Two steps of the proof cannot be omitted, and almost everywhere convergence must not be confused with convergence in L^p : (a) Minkowski's inequality and the monotone convergence theorem show that $G = \sum |f_k| \in L^p$, so that the series converges absolutely a.e. and the limit function is defined at all; (b) the dominated convergence theorem upgrades a.e. convergence to L^p convergence. If one starts from a Cauchy sequence, one also needs the fact that a Cauchy sequence with a convergent subsequence converges; this is contained in the proof of Theorem 1.5.

COROLLARY 4.3

ℓ^p ($1 \leq p < \infty$) is a Banach space; for every measure space, $L^2(X, \mu)$ is a Hilbert space.

The function G in the proof yields another useful result.

THEOREM 4.4 *An a.e. convergent subsequence*

Let $1 \leq p < \infty$ and $f_n \rightarrow f$ in L^p . Then there are a subsequence (f_{n_k}) and $h \in L^p$ such that $f_{n_k} \rightarrow f$ a.e. and $|f_{n_k}| \leq h$ a.e. for all k .

Proof. Choose $n_1 < n_2 < \dots$ with $\|f_{n_k} - f\|_p \leq 2^{-k}$ and put $G = \sum_k |f_{n_k} - f|$. As in the proof of Theorem 4.1, $\|G\|_p \leq \sum_k 2^{-k} = 1$, so $G < \infty$ a.e., and the terms of a convergent series tend to 0: $|f_{n_k} - f| \rightarrow 0$ a.e. Moreover $|f_{n_k}| \leq |f| + G =: h \in L^p$. $\square$

The next two examples show that, even on a finite measure space, convergence in L^p and a.e. convergence do not imply each other; so passing to a subsequence in Theorem 4.4 is necessary.

EXAMPLE 4.5 *The typewriter sequence*

On $[0, 1]$, for $n = 2^j + i$ ($j \geq 0, 0 \leq i < 2^j$) let $f_n = \mathbf{1}_{[i2^{-j}, (i+1)2^{-j}]}$. Then $\|f_n\|_p^p = 2^{-j} \rightarrow 0$, so $f_n \rightarrow 0$ in L^p for every $1 \leq p < \infty$. But for each $x \in [0, 1]$, the 2^j intervals of generation j cover $[0, 1]$, so $f_n(x) = 1$ for infinitely many n ; and for $j \geq 2$, x belongs to at most two of the intervals of generation j , so $f_n(x) = 0$ for infinitely many n as well. Hence $(f_n(x))$ diverges at every point. On the other hand, the subsequence $f_{2^j} = \mathbf{1}_{[0, 2^{-j}]}$ tends to 0 on $(0, 1]$, in accordance with Theorem 4.4.

EXAMPLE 4.6

On $[0, 1]$ let $g_n = n^{1/p} \mathbf{1}_{(0, 1/n)}$. Then $g_n \rightarrow 0$ everywhere, but $\|g_n\|_p = 1$, so g_n does not converge to 0 in L^p .

Almost everywhere convergence together with an L^p dominating function implies L^p convergence (dominated convergence). The exact characterisation of when a.e. convergence (or convergence in measure) can be upgraded to L^1 convergence is Vitali's convergence theorem, whose key hypothesis is uniform integrability; see the supplementary article *Uniform Integrability and L^1 Convergence*. Here is another frequently used sufficient condition.

PROPOSITION 4.7

Let $1 \leq p < \infty$, $f_n, f \in L^p$, $f_n \rightarrow f$ a.e., and $\|f_n\|_p \rightarrow \|f\|_p$. Then $\|f_n - f\|_p \rightarrow 0$.

Proof. By Proposition 2.6, $\Phi_n = 2^{p-1}(|f_n|^p + |f|^p) - |f_n - f|^p \geq 0$, and $\Phi_n \rightarrow 2^p |f|^p$ a.e. By Fatou's lemma,

$$2^p \int |f|^p \leq \liminf_{n \rightarrow \infty} \int \Phi_n = 2^p \int |f|^p - \limsup_{n \rightarrow \infty} \int |f_n - f|^p,$$

where we used $\int |f_n|^p \rightarrow \int |f|^p$. Since $\int |f|^p < \infty$, we get $\limsup \int |f_n - f|^p \leq 0$. $\square$

5 The Space L^∞

5.1 Properties of the essential supremum

PROPOSITION 5.1

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{K}$ continuous. Then $\|f\|_{L^\infty(U)} = \sup_U |f|$.

Proof. “ $\leq$ ” is clear. Let $t < \sup_U |f|$. Then $\{x \in U : |f(x)| > t\}$ is a nonempty open set; it contains an open ball and so has positive measure. Hence no $M \leq t$ is an a.e. upper bound of $|f|$, and $\|f\|_\infty \geq t$. $\square$

In particular, continuous functions that agree a.e. agree everywhere, so $f \mapsto [f]$ embeds $(C[a, b], \|\cdot\|_u)$ isometrically into $L^\infty[a, b]$ (apply the proposition on (a, b) and note $\sup_{(a,b)} |f| = \max_{[a,b]} |f|$).

5.2 Completeness

THEOREM 5.2

- (i) $f_n \rightarrow f$ in L^∞ if and only if there is a null set N such that $f_n \rightarrow f$ uniformly on $X \setminus N$.
- (ii) $L^\infty(X, \mu)$ is a Banach space.

Proof. (i) If $\|f_n - f\|_\infty \rightarrow 0$, let $N = \bigcup_n \{|f_n - f| > \|f_n - f\|_\infty\}$; by Lemma 2.3 this is a countable union of null sets, hence null, and on $X \setminus N$ we have $\sup |f_n - f| \leq \|f_n - f\|_\infty \rightarrow 0$. Conversely, if $f_n \rightarrow f$ uniformly on $X \setminus N$, then $\|f_n - f\|_\infty \leq \sup_{X \setminus N} |f_n - f| \rightarrow 0$.

(ii) Let (f_n) be a Cauchy sequence in L^∞ and put

$$N = \bigcup_{m,n} \{|f_m - f_n| > \|f_m - f_n\|_\infty\} \cup \bigcup_n \{|f_n| > \|f_n\|_\infty\},$$

a countable union of null sets, so $\mu(N) = 0$. On $X \setminus N$, $\sup |f_m - f_n| \leq \|f_m - f_n\|_\infty$, so (f_n) is uniformly Cauchy on $X \setminus N$. Let $f = \lim f_n$ on $X \setminus N$ and $f = 0$ on N ; f is measurable. Given $\varepsilon > 0$, choose N_ε with $\|f_m - f_n\|_\infty < \varepsilon$ for $m, n \geq N_\varepsilon$; letting $m \rightarrow \infty$ gives $\sup_{X \setminus N} |f_n - f| \leq \varepsilon$ for $n \geq N_\varepsilon$. Hence $|f| \leq \|f_{N_\varepsilon}\|_\infty + \varepsilon$ on $X \setminus N$, so $f \in L^\infty$, and $\|f_n - f\|_\infty \leq \varepsilon$. $\square$

REMARK 5.3

The key point of the proof is that each inequality $|f_m - f_n| \leq \|f_m - f_n\|_\infty$ holds only outside a null set, but there are only countably many index pairs (m, n) , so all exceptional sets can be collected into a single null set N , off which genuine uniform convergence takes place. Countability is essential here.

Hence ℓ^∞ is a Banach space. Together with Theorem 4.1, $L^p(X, \mu)$ is a Banach space for every $1 \leq p \leq \infty$.

5.3 The limit of $\|f\|_p$ as $p \rightarrow \infty$

THEOREM 5.4

Suppose $f \in L^r$ for some $0 < r < \infty$. Then $\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty$ (in $[0, \infty]$).

Proof. If $f = 0$ a.e. the claim is trivial, so let $\|f\|_\infty > 0$.

Lower bound: let $0 < t < \|f\|_\infty$ and $A_t = \{|f| > t\}$. By definition of the essential supremum $\mu(A_t) > 0$; by Chebyshev's inequality $\mu(A_t) \leq t^{-r} \|f\|_r^r < \infty$. Hence

$$\|f\|_p \geq \left(\int_{A_t} t^p d\mu \right)^{1/p} = t \mu(A_t)^{1/p} \xrightarrow{p \rightarrow \infty} t,$$

so $\liminf_{p \rightarrow \infty} \|f\|_p \geq t$; letting $t \uparrow \|f\|_\infty$ gives $\liminf_{p \rightarrow \infty} \|f\|_p \geq \|f\|_\infty$.

Upper bound: if $\|f\|_\infty = \infty$ we are done. Otherwise, for $p > r$, $|f|^p = |f|^{p-r} |f|^r \leq \|f\|_\infty^{p-r} |f|^r$ a.e., so

$$\|f\|_p \leq \|f\|_\infty^{1-r/p} \|f\|_r^{r/p} \xrightarrow{p \rightarrow \infty} \|f\|_\infty,$$

where we used $0 < \|f\|_r < \infty$. □

If $\mu(X) < \infty$ then $L^\infty \subseteq L^r$ (Theorem 6.1), so the theorem applies to every $f \in L^\infty$. In general the hypothesis $f \in L^r$ cannot be dropped: $f \equiv 1$ on $\mathbb{R}$ has $\|f\|_p = \infty$ for $p < \infty$ but $\|f\|_\infty = 1$.

5.4 Non-separability

A metric space is **separable** if it has a countable dense subset.

PROPOSITION 5.5

$L^\infty[0, 1]$ and ℓ^∞ are not separable.

Proof. For $t \in (0, 1]$ let $f_t = \mathbf{1}_{[0, t]}$. If $s < t$, then $|f_s - f_t| = 1$ on the set $(s, t]$ of positive measure, so $\|f_s - f_t\|_\infty = 1$. Thus the open balls $B(f_t, \frac{1}{2})$, $t \in (0, 1]$, are pairwise disjoint and uncountably many; a dense subset must meet each of them and is therefore uncountable. For ℓ^∞ consider the indicator sequences $\mathbf{1}_A$ of subsets $A \subseteq \mathbb{N}$: there are uncountably many, at mutual distance 1. □

6 Inclusions between L^p Spaces

6.1 Finite measure spaces

THEOREM 6.1

Let $\mu(X) < \infty$ and $0 < p < q \leq \infty$. Then $L^q(X, \mu) \subseteq L^p(X, \mu)$ and

$$\|f\|_p \leq \mu(X)^{\frac{1}{p} - \frac{1}{q}} \|f\|_q.$$

In particular, for $1 \leq p < q \leq \infty$ the inclusion $L^q \hookrightarrow L^p$ is a bounded linear map with operator norm $\mu(X)^{1/p-1/q}$ (attained at constant functions).

Proof. $q = \infty$: $\int |f|^p \leq \|f\|_\infty^p \mu(X)$. $q < \infty$: apply Hölder's inequality to $|f|^p$ and 1 with the exponent $r = \frac{q}{p} > 1$ and its conjugate $r' = \frac{q}{q-p}$:

$$\int |f|^p \cdot 1 \, d\mu \leq \left(\int |f|^q \, d\mu \right)^{p/q} \mu(X)^{1-p/q}.$$

Take p -th roots. For the constant function $f \equiv 1$, $\|1\|_p = \mu(X)^{1/p}$ and the two sides agree. $\square$

On a probability space ($\mu(X) = 1$), $\|f\|_p$ is nondecreasing in p , and $L^\infty \subseteq L^q \subseteq L^p \subseteq L^1$ for $1 \leq p < q \leq \infty$.

REMARK 6.2 *On the closed graph theorem*

Continuity of the inclusion can also be obtained from the closed graph theorem, which states: if V, W are Banach spaces and $T : V \rightarrow W$ is a linear map defined on *all* of V whose graph $\{(x, Tx) : x \in V\}$ is closed in $V \times W$, then T is bounded. The argument runs as follows: suppose it is known that $L^q \subseteq L^p$ ($1 \leq p, q \leq \infty$), and let $f_n \rightarrow f$ in L^q and $f_n \rightarrow g$ in L^p . By Theorem 4.4 (or Theorem 5.2 if p or q is ∞) a subsequence converges a.e. to f , and a further subsequence converges a.e. to g , so $f = g$ a.e.; the graph is closed and the inclusion is bounded. But this argument presupposes the inclusion itself, which fails on general measure spaces (Example 6.3). When $\mu(X) < \infty$, Hölder's inequality gives an explicit constant and the closed graph theorem is not needed.

6.2 No inclusions in general

EXAMPLE 6.3

The finiteness assumption in Theorem 6.1 cannot be dropped. On $(\mathbb{R}, m)$, let $0 < p < q < \infty$. By Example 2.10,

$$x^{-1/p} \mathbf{1}_{(1,\infty)} \in L^q \setminus L^p, \quad x^{-1/q} \mathbf{1}_{(0,1)} \in L^p \setminus L^q.$$

(For the first, $\frac{q}{p} > 1$ while $\frac{p}{p} = 1$; for the second, $\frac{p}{q} < 1$ while $\frac{q}{q} = 1$.) Also $1 \in L^\infty(\mathbb{R}) \setminus L^p(\mathbb{R})$ and $x^{-1/(2p)} \mathbf{1}_{(0,1)} \in L^p(\mathbb{R}) \setminus L^\infty(\mathbb{R})$. So on $\mathbb{R}$ no two L^p spaces with different exponents are comparable.

6.3 Sequence spaces: the inclusion reverses

THEOREM 6.4

Let $0 < p < q \leq \infty$. Then $\ell^p \subseteq \ell^q$ and $\|x\|_q \leq \|x\|_p$. The inclusion is strict.

Proof. $q = \infty$: for each n , $|x_n|^p \leq \sum_k |x_k|^p$. Let $q < \infty$ and $x \neq 0$. By homogeneity we may

assume $\|x\|_p = 1$; then $|x_n| \leq 1$, hence $|x_n|^q \leq |x_n|^p$, and summing gives $\|x\|_q^q \leq 1$. Strictness: $x_n = n^{-1/p}$ satisfies $\sum n^{-q/p} < \infty$ but $\sum n^{-1} = \infty$, so $x \in \ell^q \setminus \ell^p$. $\square$

Compare Example 2.10: a finite measure rules out “largeness at infinity”, so spaces with smaller exponents are larger; under counting measure every nonempty set has measure at least 1, which rules out “local singularities”, so spaces with larger exponents are larger. On $\mathbb{R}$ both phenomena are present, and there is no inclusion at all.

6.4 The interpolation inequality

THEOREM 6.5 *Interpolation inequality*

Let $0 < p < r < q \leq \infty$ and define $\theta \in (0, 1)$ by

$$\frac{1}{r} = \frac{\theta}{p} + \frac{1-\theta}{q}.$$

Then for $f \in L^p \cap L^q$,

$$\|f\|_r \leq \|f\|_p^\theta \|f\|_q^{1-\theta}.$$

In particular $L^p \cap L^q \subseteq L^r$.

Proof. If $q = \infty$, then $\theta = \frac{p}{r}$, and $|f|^r = |f|^p |f|^{r-p} \leq \|f\|_\infty^{r-p} |f|^p$ a.e.; integrating and taking r -th roots gives $\|f\|_r \leq \|f\|_p^{p/r} \|f\|_\infty^{1-p/r}$. Let $q < \infty$. Write $|f| = |f|^\theta \cdot |f|^{1-\theta}$ and apply the generalised Hölder inequality (Corollary 3.6) with exponents $\frac{p}{\theta}$ and $\frac{q}{1-\theta}$, which is legitimate precisely because $\frac{1}{r} = \frac{\theta}{p} + \frac{1-\theta}{q}$:

$$\|f\|_r \leq \| |f|^\theta \|_{p/\theta} \| |f|^{1-\theta} \|_{q/(1-\theta)} = \|f\|_p^\theta \|f\|_q^{1-\theta}.$$

$\square$

REMARK 6.6

The interpolation inequality says equivalently that $\frac{1}{p} \mapsto \log \|f\|_p$ is a convex function on the interval where it is finite. The analogous statement for linear operators is the Riesz–Thorin interpolation theorem, which we do not pursue here.

COROLLARY 6.7

Let $0 < p < r < q \leq \infty$. Then $L^r \subseteq L^p + L^q$, i.e. every $f \in L^r$ can be written as $f = g + h$ with $g \in L^p$ and $h \in L^q$.

Proof. Let $g = f \mathbf{1}_{\{|f| > 1\}}$ and $h = f \mathbf{1}_{\{|f| \leq 1\}}$. On $\{|f| > 1\}$ we have $|f|^p \leq |f|^r$, so $g \in L^p$. If $q = \infty$, then $|h| \leq 1$; if $q < \infty$, then $|f|^q \leq |f|^r$ on $\{|f| \leq 1\}$, so $h \in L^q$. $\square$

Intuitively, the “tall peaks” of f belong to the space with the smaller exponent and the “low plateau” to the space with the larger exponent.

7 Dense Subspaces and Separability

7.1 Simple functions

Recall the approximation theorem of Article II: for a nonnegative measurable $f : X \rightarrow [0, \infty]$ there are simple functions $0 \leq \varphi_1 \leq \varphi_2 \leq \dots$ converging to f pointwise, and uniformly on every set on which f is bounded. For a $\mathbb{K}$ -valued measurable $f = u + iv$, apply this to $u^\pm$ and $v^\pm$ to obtain simple functions $\varphi_n = (a_n - b_n) + i(c_n - d_n)$ with $0 \leq a_n \uparrow u^+$, etc. Since a_n and b_n have disjoint supports, $|a_n - b_n| = a_n + b_n \leq |u|$, and likewise $|c_n - d_n| \leq |v|$; hence

$$|\varphi_n| \leq |f|, \quad \varphi_n \rightarrow f \text{ pointwise,}$$

uniformly on every set on which f is bounded.

THEOREM 7.1

- (i) Let $1 \leq p < \infty$. The simple functions of the form $\sum_{i=1}^N a_i \mathbf{1}_{E_i}$ ($a_i \in \mathbb{K}$, $\mu(E_i) < \infty$) are dense in $L^p(X, \mu)$.
- (ii) The simple functions are dense in $L^\infty(X, \mu)$.

Proof. (i) Let $f \in L^p$ and take φ_n as above. If φ_n takes a nonzero value a on a set E , then $|f| \geq |a|$ on E , and Chebyshev's inequality gives $\mu(E) \leq |a|^{-p} \|f\|_p^p < \infty$; so φ_n is of the stated form. Moreover $|f - \varphi_n|^p \leq (2|f|)^p \in L^1$ and $|f - \varphi_n|^p \rightarrow 0$ pointwise, so $\|f - \varphi_n\|_p \rightarrow 0$ by dominated convergence.

(ii) Let $f \in L^\infty$. Redefining f to be 0 on the null set $\{|f| > \|f\|_\infty\}$ we obtain a representative that is bounded everywhere; then $\varphi_n \rightarrow f$ uniformly on X , so $\|f - \varphi_n\|_\infty \leq \sup_X |f - \varphi_n| \rightarrow 0$. $\square$

REMARK 7.2

In (i) the condition $\mu(E_i) < \infty$ is automatic: a simple function in L^p ($p < \infty$) has finite-measure level sets for its nonzero values. In (ii) one cannot require $\mu(E_i) < \infty$: if $\mu(X) = \infty$, every such simple function φ vanishes on a set of infinite measure, so $\|1 - \varphi\|_\infty \geq 1$.

7.2 Continuous functions with compact support on $\mathbb{R}^n$

Let $C_c(\mathbb{R}^n)$ be the space of continuous $\mathbb{K}$ -valued functions on $\mathbb{R}^n$ whose support $\text{supp } f = \overline{\{f \neq 0\}}$ is compact. Such functions are bounded and vanish outside a set of finite measure, so $C_c(\mathbb{R}^n) \subseteq L^p(\mathbb{R}^n)$ for all $0 < p \leq \infty$.

LEMMA 7.3 Urysohn's lemma in $\mathbb{R}^n$

Let $K \subseteq U \subseteq \mathbb{R}^n$ with K compact and U open. Then there is $g \in C_c(\mathbb{R}^n)$ with $0 \leq g \leq 1$, $g = 1$ on K and $\text{supp } g \subseteq U$.

Proof. If $U = \mathbb{R}^n$ let $\delta = 1$; otherwise let $\delta = \inf_{x \in K} d(x, U^c)$. The function $x \mapsto d(x, U^c)$ is continuous and positive on K and attains its minimum on the compact set K , so $\delta > 0$ (if $K = \emptyset$ take $g = 0$). Put

$$g(x) = \max\left(0, 1 - \frac{2d(x, K)}{\delta}\right).$$

Since $d(\cdot, K)$ is 1-Lipschitz, g is continuous, $0 \leq g \leq 1$ and $g = 1$ on K . Moreover $\text{supp } g \subseteq \{x : d(x, K) \leq \delta/2\}$, a closed bounded set, hence compact; its points are at distance less than δ from K , so they cannot lie in U^c , and it is contained in U . $\square$

THEOREM 7.4

For $1 \leq p < \infty$, $C_c(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$.

Proof. By Theorem 7.1(i) and Minkowski's inequality it suffices to show: for every measurable E with $m(E) < \infty$ and every $\varepsilon > 0$ there is $g \in C_c(\mathbb{R}^n)$ with $\|\mathbf{1}_E - g\|_p < \varepsilon$. (If $\|\mathbf{1}_{E_i} - g_i\|_p < \varepsilon$, then $\|\sum a_i \mathbf{1}_{E_i} - \sum a_i g_i\|_p < \varepsilon \sum |a_i|$.) By regularity of Lebesgue measure (Article I) there are a compact K and an open U with $K \subseteq E \subseteq U$ and $m(U \setminus K) < \varepsilon^p$. Take g as in Lemma 7.3; then $\mathbf{1}_K \leq g \leq \mathbf{1}_U$, and hence $|g - \mathbf{1}_E| \leq \mathbf{1}_{U \setminus K}$: on K both equal 1, outside U both vanish, and on $U \setminus K$ both lie in $[0, 1]$. Therefore

$$\|g - \mathbf{1}_E\|_p \leq m(U \setminus K)^{1/p} < \varepsilon.$$

$\square$

REMARK 7.5

- (i) The general form of this result is: if X is a locally compact Hausdorff space and μ a Radon measure on X , then $C_c(X)$ is dense in $L^p(X, \mu)$ for $1 \leq p < \infty$. We have proved only the case of $\mathbb{R}^n$; the same proof works for $C_c(U)$ in $L^p(U)$, $U \subseteq \mathbb{R}^n$ open.
- (ii) The result fails for $p = \infty$. In fact the L^∞ -distance from $\mathbf{1}_{[0,1]}$ to $C_c(\mathbb{R})$ is at least $\frac{1}{2}$: if g is continuous with $\|g - \mathbf{1}_{[0,1]}\|_\infty = c < \frac{1}{2}$, then $\{x \in (0, 1) : |g(x) - 1| > c\}$ is an open null set, hence empty, i.e. $|g - 1| \leq c$ on $(0, 1)$, and by continuity $g(1) \geq 1 - c > \frac{1}{2}$; likewise $|g| \leq c$ on $(1, 2)$ gives $g(1) \leq c < \frac{1}{2}$, a contradiction.
- (iii) By Theorems 4.1 and 7.4, $L^p(\mathbb{R}^n)$ is a Banach space containing $C_c(\mathbb{R}^n)$ as a dense subspace, i.e. it is the completion of $(C_c(\mathbb{R}^n), \|\cdot\|_p)$. Likewise $L^1[0, 2]$ is the completion of the incomplete space $(C[0, 2], \|\cdot\|_1)$ of Example 1.13.

As a typical application of density we show that translation is continuous in L^p . For $h \in \mathbb{R}^n$ let $(\tau_h f)(x) = f(x - h)$. By translation invariance of Lebesgue measure (Article I), $\int f(x - h) dx = \int f(x) dx$ for indicator functions, hence for simple functions, and by monotone convergence for nonnegative measurable functions; therefore $\|\tau_h f\|_p = \|f\|_p$.

THEOREM 7.6 Continuity of translation

Let $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^n)$. Then $\lim_{h \rightarrow 0} \|\tau_h f - f\|_p = 0$.

Proof. Given $\varepsilon > 0$, choose by Theorem 7.4 a function $g \in C_c(\mathbb{R}^n)$ with $\|f - g\|_p < \varepsilon$, and let $\text{supp } g \subseteq B(0, R)$. Being continuous on a compact set and zero outside it, g is uniformly continuous on $\mathbb{R}^n$, so $\sup_x |g(x - h) - g(x)| \rightarrow 0$ as $h \rightarrow 0$. For $|h| \leq 1$, $\tau_h g - g$ vanishes outside $B(0, R + 1)$, hence

$$\|\tau_h g - g\|_p \leq \sup_x |g(x - h) - g(x)| \cdot m(B(0, R + 1))^{1/p} \xrightarrow{h \rightarrow 0} 0.$$

Therefore

$$\|\tau_h f - f\|_p \leq \|\tau_h(f - g)\|_p + \|\tau_h g - g\|_p + \|g - f\|_p < 2\varepsilon + \|\tau_h g - g\|_p,$$

and $\limsup_{h \rightarrow 0} \|\tau_h f - f\|_p \leq 2\varepsilon$. $\square$

The result fails for $p = \infty$: for $f = \mathbf{1}_{[0,1]}$ and any $0 < |h| < 1$, $\tau_h f - f$ takes the values ± 1 on a set of positive measure, so $\|\tau_h f - f\|_\infty = 1$.

7.3 Separability

THEOREM 7.7

For $1 \leq p < \infty$, $L^p(\mathbb{R}^n)$ is separable.

Proof. For $j \geq 0$, a dyadic cube of order j is a set $\prod_{i=1}^n [k_i 2^{-j}, (k_i + 1) 2^{-j}]$ with $k \in \mathbb{Z}^n$. Let $\mathcal{D}$ be the set of finite linear combinations, with coefficients in $\mathbb{Q} + i\mathbb{Q}$ (in $\mathbb{Q}$ in the real case), of indicator functions of dyadic cubes; $\mathcal{D}$ is countable.

Given $f \in L^p$ and $\varepsilon > 0$, choose $g \in C_c(\mathbb{R}^n)$ with $\|f - g\|_p < \varepsilon$ and a positive integer R with $\text{supp } g \subseteq Q_R = [-R, R]^n$. By uniform continuity of g there is j such that $|g(x) - g(y)| < \eta$ whenever x, y lie in the same dyadic cube of order j , where $\eta > 0$ is to be chosen. Q_R is tiled exactly by finitely many dyadic cubes Q of order j ; for each such Q choose a point $x_Q \in Q$ and $c_Q \in \mathbb{Q} + i\mathbb{Q}$ with $|c_Q - g(x_Q)| < \eta$, and put $\psi = \sum_Q c_Q \mathbf{1}_Q \in \mathcal{D}$. Then $|g - \psi| < 2\eta$ on Q_R and $g = \psi = 0$ outside Q_R , so

$$\|g - \psi\|_p \leq 2\eta (2R)^{n/p}.$$

Choosing η so that the right side is less than ε gives $\|f - \psi\|_p < 2\varepsilon$. $\square$

REMARK 7.8

Separability is not a property of L^p ($1 \leq p < \infty$) on general measure spaces. Let X be an uncountable set (e.g. $\mathbb{R}$) with counting measure; the functions $e_x = \mathbf{1}_{\{x\}}$ ($x \in X$) satisfy $\|e_x - e_y\|_p = 2^{1/p}$ for $x \neq y$, and the argument of Proposition 5.5 shows that $L^p(X, \mu)$ is not separable. A commonly used sufficient condition is that μ be σ -finite and $\mathcal{M}$ be countably generated (modulo null sets); then $L^p(X, \mu)$ is separable for $1 \leq p < \infty$. L^∞ , on the other hand, fails to be separable in essentially every infinite situation of interest (Proposition 5.5).

8 A Brief Introduction to Duality

This section is only a brief discussion; the full representation theorem requires the Radon–Nikodym theorem and is postponed.

PROPOSITION 8.1

Let $1 \leq p \leq \infty$, let q be the conjugate exponent and $g \in L^q(X, \mu)$. Then

$$\Lambda_g(f) = \int_X f g \, d\mu \quad (f \in L^p)$$

defines a bounded linear functional on L^p with $\|\Lambda_g\| \leq \|g\|_q$. If $1 < p \leq \infty$, or if $p = 1$ and μ is σ -finite, then $\|\Lambda_g\| = \|g\|_q$. Thus in these cases $g \mapsto \Lambda_g$ is a linear isometry (in

particular injective) from L^q into $(L^p)^*$.

Proof. $\Lambda_g(f)$ does not depend on the representatives, is linear in f , and $|\Lambda_g(f)| \leq \|f\|_p \|g\|_q$ by Hölder's inequality. We prove the reverse inequality, assuming $g \neq 0$. Let $\operatorname{sgn} z = z/|z|$ for $z \neq 0$ and $\operatorname{sgn} 0 = 0$; then $g \overline{\operatorname{sgn} g} = |g|$.

$1 < p < \infty$: let $f = |g|^{q-1} \overline{\operatorname{sgn} g}$. Then $|f|^p = |g|^{(q-1)p} = |g|^q$, so $\|f\|_p = \|g\|_q^{q/p} < \infty$, and $\Lambda_g(f) = \int |g|^q = \|g\|_q^q$. Hence

$$\|\Lambda_g\| \geq \frac{\|g\|_q^q}{\|g\|_q^{q/p}} = \|g\|_q^{q(1-1/p)} = \|g\|_q.$$

$p = \infty$ ($q = 1$): let $f = \overline{\operatorname{sgn} g}$; then $\|f\|_\infty \leq 1$ and $\Lambda_g(f) = \|g\|_1$.

$p = 1$ ($q = \infty$), μ σ -finite: let $0 < t < \|g\|_\infty$; then $A = \{|g| > t\}$ has $\mu(A) > 0$. Write $X = \bigcup_k X_k$ with X_k increasing and $\mu(X_k) < \infty$; then $\mu(A \cap X_k) \uparrow \mu(A) > 0$, so some $B = A \cap X_k$ has $0 < \mu(B) < \infty$. Let $f = \mu(B)^{-1} \mathbf{1}_B \overline{\operatorname{sgn} g}$; then $\|f\|_1 = 1$ and

$$\Lambda_g(f) = \frac{1}{\mu(B)} \int_B |g| d\mu \geq t.$$

So $\|\Lambda_g\| \geq t$; let $t \uparrow \|g\|_\infty$. □

THEOREM 8.2 *Riesz representation theorem for L^p*

- (i) Let $1 < p < \infty$. For every $\Lambda \in (L^p)^*$ there is a unique $g \in L^q$ with $\Lambda = \Lambda_g$; hence $(L^p)^*$ is isometrically isomorphic to L^q .
- (ii) Let μ be σ -finite. Then $(L^1)^*$ is isometrically isomorphic to L^∞ (again via $g \mapsto \Lambda_g$).

The proof requires the Radon–Nikodym theorem (or, for (i), the uniform convexity of L^p) and is omitted here. Uniqueness and the isometry property are already given by Proposition 8.1; the difficulty lies in surjectivity. Note that the representing function g belongs to L^q , not to L^p .

EXAMPLE 8.3 *σ -finiteness cannot be omitted*

Let $X = \{a\}$ with $\mu(\{a\}) = \infty$. Then $\int |f| d\mu = |f(a)| \cdot \infty$, so $L^1 = \{0\}$ and $(L^1)^* = \{0\}$; but X has no nonempty null sets, so $L^\infty = \mathbb{K}$. The map $g \mapsto \Lambda_g$ is the zero map and is not injective.

REMARK 8.4 *The case $p = \infty$*

By Proposition 8.1, L^1 embeds isometrically into $(L^\infty)^*$, but in general not surjectively, so “ $(L^p)^* = L^q$ ” fails in general for $p = \infty$. Take ℓ^∞ : on the subspace c of convergent sequences, $\ell(x) = \lim_n x_n$ is a bounded linear functional of norm 1; by the Hahn–Banach theorem it extends with the same norm to a functional Λ on ℓ^∞ . If $\Lambda = \Lambda_y$ with $y \in \ell^1$, then $y_k = \Lambda(e_k) = \lim_n (e_k)_n = 0$ for every k , so $\Lambda = 0$, contradicting $\Lambda(1, 1, \dots) = 1$.

REMARK 8.5

When μ is σ -finite, $L^\infty = (L^1)^*$, and by the Banach–Alaoglu theorem the closed unit ball of L^∞ is compact in the weak* topology $\sigma(L^\infty, L^1)$. For $1 < p < \infty$, Theorem 8.2 gives $(L^p)^{**} \cong (L^q)^* \cong L^p$, i.e. L^p is reflexive. These matters belong to functional analysis.

Summary

The table summarises the main results of this article on $L^p(X, \mu)$ (“ $\mathbb{R}^n$ ” indicates a statement proved for $(\mathbb{R}^n, m)$ that may fail on general measure spaces). In the column $0 < p < 1$, rows 2–5 follow by the same proofs as for $p \geq 1$, with the metric d_p in place of the norm.

Property	$0 < p < 1$	$1 \leq p < \infty$	$p = \infty$
$\ \cdot\ _p$ is a norm	no	yes	yes
complete	yes (metric d_p)	yes	yes
simple functions dense	yes	yes	yes
C_c dense	yes ($\mathbb{R}^n$)	yes ($\mathbb{R}^n$)	no
separable	yes ($\mathbb{R}^n$)	yes ($\mathbb{R}^n$)	no
dual	trivial ($\mathbb{R}^n$)	L^q ($p = 1$: σ -finite)	in general $\supsetneq L^1$

The proofs rest on only three tools. Young’s inequality (a consequence of convexity) yields Hölder’s inequality, and Hölder’s inequality yields Minkowski’s; Minkowski’s inequality together with the monotone and dominated convergence theorems gives completeness; and the regularity of Lebesgue measure gives the density of C_c and separability. The relations between different L^p spaces are governed entirely by the behaviour of the measure “locally” and “at infinity”.

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